Ch. 10 CW

Principles of Microeconomics

October 27, 2025

Instructions

Please justify all of your answers and submit your solutions on Gradescope by the beginning of next class.

Problems

1. What type of externality (positive or negative) is present in each of the following examples? Is the social value of the activity greater than or equal to the private value to the individual? Is the social cost of the activity greater than or equal to the private cost to the individual? Without intervention, will there be too little or too much (relative to what would be efficient) of this activity?

   (a) Mr. Chau plants lots of colorful flowers in his front yard.

   (b) Your next-door neighbor likes to build bonfires in his backyard, and sparks often drift onto your house.

   (c) Maija, who lives next to an apple orchard, decides to keep bees to produce honey.

   (d) Justine buys a large SUV that consumes a lot of gasoline.

2. Suppose that tanneries pollute nearby bodies of water. The supply and demand for leather goods are given by $Q_S = \frac{1}{2}P$ and $Q_D = 10 - 2P$. If the pollution creates a \$3 per unit externality, please answer the following.

   (a) What are the price and quantity at the market equilibrium?

   (b) What are the price and quantity at the efficient equilibrium?

   (c) What is deadweight loss at the market equilibrium?

3. Supply for solar cells is given by $Q_S = 4P$, and demand is given by $Q_D = 14 - 2P$. Because they reduce carbon emissions, every solar cell sold generates \$2 worth of external value.

   (a) What quantity of solar cells is sold at the market equilibrium?

   (b) What quantity of solar cells is sold at the efficient equilibrium?

   (c) What government policy would generate the efficient equilibrium?

4. Suppose Harmon has a dog that he loves, but its barking disturbs Andy. Assume Harmon and Andy live together, so they can negotiate at zero cost.

   (a) Harmon values having his dog at $10,000, but Andy values his quiet at $20,000. What's the efficient outcome – Harmon keeping the dog or ditching the dog?

   (b) If Harmon has a right to keep his dog, what would result from Andy and Harmon negotiating?

   (c) If Andy has a right to a quiet apartment, what would result from Andy and Harmon negotiating?

   (d) What effect does the distribution of rights have on the outcomes in (b) and (c)?

5. The two dry-cleaning companies in Collegetown, College Cleaners and Big Green Cleaners, are a major source of air pollution. Together they currently produce 350 units of air pollution, which the town wants to reduce to 200 units. They accompanying table shows the current pollution level produced by each company and each company's cost of reducing its pollution.

Companies	Initial Pollution Level	Cost of Reducing Pollution (per unit)
College Cleaners	230 units	$5
Big Green Cleaners	120 units	$2

   (a) Suppose that Collegetown passes an environmental standards law that limits each company to 100 units of pollution. What would be the total cost to the two companies of each reducing its pollution emissions to 100 units?

   (b) Suppose instead that Collegetown issues 100 pollution vouchers to each company, each entitling the company to one unit of pollution, and that these vouchers can be traded. How much is each pollution voucher worth to College Cleaners? To Big Green Cleaners? (That is, how much would each company, at most, be willing to pay for one more voucher?)

   (c) Who will sell vouchers and who will buy them? How many vouchers will be traded?

   (d) What is the total cost to the two companies of the pollution controls under this voucher system?